

Robustness of Hebrew Sentiment Analysis to Meaning-Preserving Orthographic Variation

Prompting vs. Fine-Tuning — Final Project (V2)

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Abstract

Hebrew is written with pervasive, meaning-preserving orthographic variation — full vs. defective spelling (*ktiv male/haser*), optional niqqud, clitic prefixes, and casual typing — that a human reads identically. We ask not which of four Hebrew sentiment methods is most *accurate*, but which is most *robust* to such variation, and why. Reusing our V1 pipeline (zero-shot, few-shot, and retrieved few-shot Claude Haiku 4.5, plus a LoRA-fine-tuned DictaBERT) on a frozen 500-example subset of HebArabNlpProject/HebrewSentiment, we build a deterministic five-family perturbation suite and measure a Robustness Gap (Macro-F1 drop) and a label-independent Flip-Rate. We find that RAG-augmented prompting is the most stable method overall, while the Hebrew-native LoRA model — invariant to spelling and niqqud — is paradoxically the most fragile to final-form typo normalization. A mechanistic tokenization analysis shows that although niqqud shatters an English-centric BPE tokenizer far more than DictaBERT's Hebrew WordPiece, prediction flips do *not* correlate with tokenization change, so tokenizer fragmentation alone does not explain fragility.

1 Introduction and Motivation

Standard NLP evaluation reports a single accuracy number on clean, canonical text. For Hebrew this is especially misleading. Hebrew orthography admits many surface forms that denote the same word and are read identically by native speakers: a word may be written with or without matres lectionis (*ktiv male* vs. *ktiv haser*, e.g. תוכנית ↔ תוכנית); niqqud (vowel points) may be present or absent; single-letter prepositions and conjunctions attach as prefixes to the following word; and everyday user-generated text is full of typos and non-standard final-letter usage. None of these change meaning, yet each changes the byte string a model sees.

A model that is accurate on clean text but changes its prediction when the same sentence is spelled differently is fragile in a way that a clean accuracy score cannot reveal. This matters directly for Hebrew NLP deployed on real user input — reviews, surveys, social media — where such variation is the norm rather than the exception. Our contribution is a controlled, *meaning-preserving* robustness evaluation that isolates surface-form sensitivity from semantic difficulty, applied uniformly to four qualitatively different methods, and paired with a mechanistic probe of *why* fragility occurs. The framing deliberately follows the course guideline that a fair, well-analysed evaluation is valuable even when a method does not “win”.

Concretely we ask three questions. **(Q1)** Which of the four methods is most robust to meaning-preserving Hebrew orthographic variation? **(Q2)** For the prompting methods, does grounding the instruction in Hebrew rather than English buy robustness? **(Q3)** Is fragility explained by how badly each tokenizer fragments the perturbed text?

2 Background and Related Work

Hebrew encoder models. DictaBERT (Shmidman et al., 2023) is a BERT encoder pre-trained on modern Hebrew with a Hebrew WordPiece vocabulary; we fine-tune it with LoRA. For retrieval we use neodictabert-bilingual-embed, a Hebrew–English sentence embedding model from the same group. Because their vocabularies are trained on Hebrew, they are expected to fragment Hebrew text far less than English-centric byte-pair tokenizers.

Sentiment data. HebArabNlpProject/HebrewSentiment is a three-class (Negative/Neutral/Positive) Hebrew sentiment dataset of survey and review text. It is naturally noisy and colloquial, which suits a robustness study.

Orthography. Hebrew spelling follows two conventions, *ktiv male* (full, with matres lectionis) and *ktiv haser* (defective); niqqud diacritics disambiguate vowels but are usually omitted. We generate niqqud with the Dicta Nakdan service and curate a high-precision spelling map. These phenomena are the linguistic basis for our perturbations.

Robustness evaluation. Perturbation-based testing (e.g. CheckList, Ribeiro et al., 2020) probes model behaviour under controlled input edits. We adapt this idea to Hebrew orthography and add a label-independent Flip-Rate, and —

as the novel mechanistic component — we relate flips to tokenizer fertility using DictaBERT WordPiece, tiktoken o200k BPE, and Claude's own measured input-token counts.

3 Methodology

3.1 The four methods (reused from V1)

- **M1 — Zero-shot Claude (Haiku 4.5).** Direct classification via the Anthropic Messages API, temperature 0, JSON-shaped label. Run with semantically identical Hebrew and English instruction templates.
- **M2 — Few-shot Claude.** Same, with nine curated demonstrations (three per class) drawn from the training split only; Hebrew and English variants.
- **M3 — Retrieved few-shot Claude (RAG).** Demonstrations retrieved per input with neodictabert-bilingual-embed over a FAISS cosine index with class-rebalanced top-k; Hebrew and English variants. Crucially, retrieval is run on the text actually presented to the model, so RAG robustness includes realistic retrieval drift.
- **M4 — LoRA DictaBERT.** dicta-il/dictabert + a 3-class head, LoRA ($r=8$, $\alpha=16$, dropout 0.1) on the attention query/value projections; only LoRA and the head are trained (297k / 184.6M params, 0.16%). 3 epochs, early stopping on validation Macro-F1.

All seven conditions (six prompting + LoRA) are scored on the same frozen 500-example subset so numbers are directly comparable. Every Claude request passes through a SHA-keyed disk cache, making reruns free; the entire robustness sweep ($\approx 18k$ Haiku calls) cost $\approx \$1$ one-time.

3.2 Meaning-preserving perturbation suite (new in V2)

Each family is a deterministic function $\text{perturb}(\text{text}) \rightarrow (\text{new_text}, \text{changed})$. The *changed* flag is essential: many edits are no-ops on a given sentence (no splittable prefix, no final letter, ...), and both metrics are computed only where the text actually changed, so coverage differences never inflate or deflate a score.

- **spelling** — curated full $\leftrightarrow$ defective rewrites (e.g. תוכנית $\leftrightarrow$ תכנית); high precision, intentionally low coverage.
- **niqqud** — niqqud insertion via the Dicta Nakdan API (cached; degrades to a no-op if offline).
- **prefix** — vocabulary-aware clitic segmentation (ב בית $\rightarrow$ בבית), only when the remainder is a real word.
- **typos** — realistic typos and final-form normalization (ג $\rightarrow$ ג, מ $\rightarrow$ מ, ...).
- **colloquial** — formal $\leftrightarrow$ colloquial paraphrase generated by Claude and *validated to preserve sentiment*; this is a semantic edit, so it is reported separately from the four rule-based families.

3.3 Metrics

- **Robustness Gap** = $\text{Macro-F1}(\text{clean}) - \text{Macro-F1}(\text{family})$, per condition $\times$ family. A small or negative gap means the method is behaviourally stable.
- **Flip-Rate** = among examples a method got correct on clean input (and where the edit changed the text), the fraction whose prediction flips. It needs no gold labels, so it isolates surface sensitivity from label difficulty.
- **Tokenizer fertility** = change in total tokens per sentence induced by an edit, under DictaBERT WordPiece, tiktoken o200k (an open BPE reference), and Claude's own per-request `input_tokens` (the model-faithful signal, since Claude's tokenizer is not public). We test whether flips coincide with larger tokenization change via point-biserial correlation.

4 Data, Models, and Reproducibility

The dataset ships 43,645 rows across three classes, heavily skewed toward Neutral (Neutral 25,033; Positive 10,049; Negative 8,563). We take a stratified 80/10/10 split (train 34,916 / val 4,364 / test 4,365, seed 42) and freeze a class-balanced 500-example subset of the test split (167/167/166) as the common evaluation set; its row IDs are persisted so every method sees exactly the same inputs. The few-shot pool (9 examples) is sampled from the training split only, avoiding any test leakage. Figure 1 shows the character-length distribution: texts are short (median 30 characters), which makes single-word orthographic edits proportionally impactful.

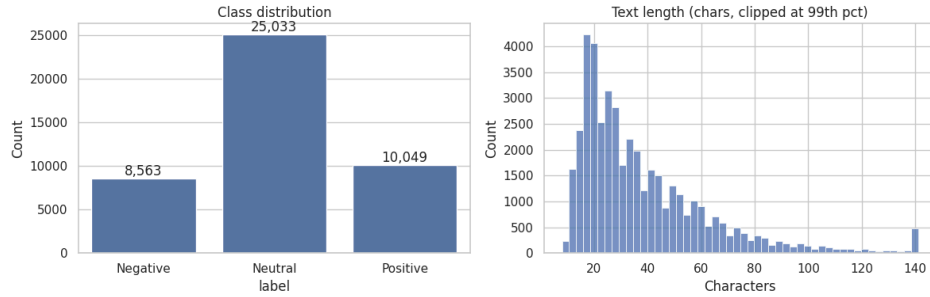


Figure 1. Character-length distribution of the HebrewSentiment corpus (median 30, mean 38.7 characters). Short texts amplify the effect of single-word orthographic edits.

Models/tools: Claude Haiku 4.5 (prompting), dicta-il/dictabert (LoRA), dicta-il/neodictabert-bilingual-embed + FAISS (RAG), Dicta Nakdan (niqqud), tiktoken and the DictaBERT tokenizer (fertility). All artifacts (frozen subset, few-shot pool, embeddings, LoRA adapter, per-request cache) are persisted and reused, so the full pipeline is deterministic and re-runnable end-to-end.

5 Results

Clean-input Macro-F1 (Table 1, first column) reproduces the familiar ordering: retrieval-augmented Claude is strongest (RAG-EN 0.858, RAG-HE 0.845), the LoRA model is competitive (0.844), and plain zero-shot is weakest (≈ 0.80). But the robustness picture is different from the accuracy picture.

Condition	Clean F1	spelling	niqqud	prefix	typos	colloq.
ZS-Claude (HE)	0.805	+0.000	0.018	0.016	0.010	0.037
ZS-Claude (EN)	0.798	0.000	0.014	0.019	0.028	0.041
FS-Claude (HE)	0.808	0.000	-0.001	0.023	0.012	0.022
FS-Claude (EN)	0.811	-0.002	0.005	0.024	0.025	0.022
RAG-Claude (HE)	0.845	-0.002	-0.008	0.004	-0.016	0.027
RAG-Claude (EN)	0.858	-0.002	+0.000	0.016	-0.004	0.041
LoRA-DictaBERT	0.844	0.000	0.000	0.021	0.056	0.023

Table 1. Robustness Gap = $\text{Macro-F1}(\text{clean}) - \text{Macro-F1}(\text{family})$. Smaller (or negative) is more robust. RAG rows are near-zero on rule-based families; LoRA’s typos gap (0.056) is the largest single orthographic gap.

Condition	spelling	niqqud	prefix	typos	colloq.
ZS-Claude (HE)	0.000	0.058	0.064	0.053	0.084
ZS-Claude (EN)	0.000	0.054	0.057	0.070	0.093
FS-Claude (HE)	0.027	0.038	0.068	0.049	0.069
FS-Claude (EN)	0.000	0.038	0.068	0.063	0.069
RAG-Claude (HE)	0.000	0.026	0.041	0.036	0.089
RAG-Claude (EN)	0.000	0.051	0.056	0.039	0.091
LoRA-DictaBERT	0.000	0.000	0.064	0.118	0.095

Table 2. Flip-Rate = fraction of clean-correct predictions that change under a meaning-preserving edit (label-independent). LoRA is invariant to spelling/niqqud but has the highest typos flip-rate (0.118).

Family coverage (fraction of the 500 examples actually changed) is spelling 8.2%, niqqud 45.8%, prefix 71.0%, typos 72.4%, colloquial 93.2%; metrics are computed only on changed examples. The heatmaps in Figure 2 make the two headline patterns visible at a glance: RAG rows are the coolest across the rule-based families, and the LoRA cell for typos is a clear hotspot in both panels.

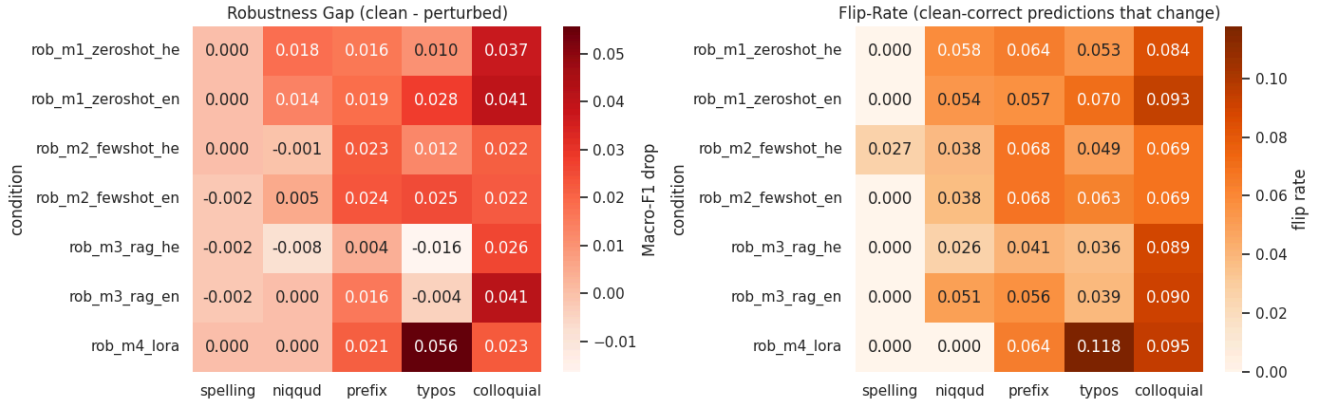


Figure 2. Robustness Gap (left) and Flip-Rate (right) across the seven conditions and five perturbation families. Cooler = more robust. RAG is stable on orthographic families; LoRA spikes on typos.

5.1 Mechanistic tokenization analysis

Table 3 reports the mean per-sentence token increase each family induces under the two open tokenizers. The signature result is niqqud: it adds **+30.1 tokens** on average under tiktoken's English-centric BPE while leaving DictaBERT's Hebrew WordPiece *completely unchanged* ($\Delta = 0$). Prefix and typo edits fragment DictaBERT slightly more than tiktoken, whereas curated spelling barely moves either.

Family	Coverage	Δ tokens (DictaBERT)	Δ tokens (tiktoken)
spelling	8.2%	0.02	0.32
niqqud	45.8%	0.00	30.08
prefix	71.0%	1.98	1.12
typos	72.4%	2.04	1.31
colloquial	93.2%	1.78	1.77

Table 3. Mean per-sentence tokenization change and coverage per family. Niqqud shatters the BPE tokenizer (+30.1) but not the Hebrew WordPiece tokenizer (0).

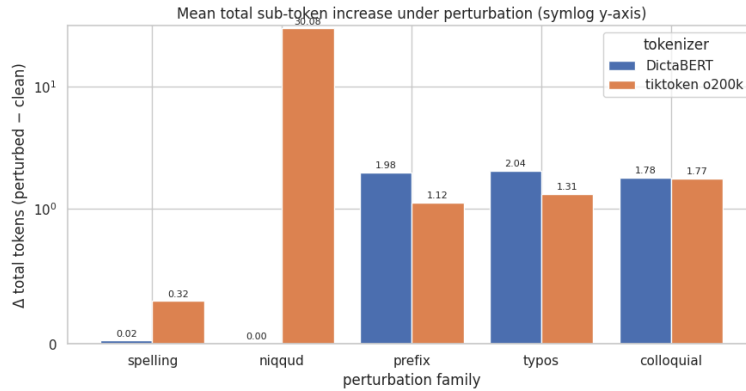


Figure 3. Tokenizer fertility change per family: an English-centric BPE fragments perturbed Hebrew (especially niqqud) far more than DictaBERT's Hebrew-trained WordPiece.

Yet the causal test fails. Point-biserial correlations between a prediction flip and the size of the tokenization change are essentially zero and non-significant: API flip vs. tiktoken $\Delta = -0.017$ ($p=0.56$); API flip vs. Claude Δ input-tokens = 0.017 ($p=0.55$); LoRA flip vs. DictaBERT $\Delta = 0.046$ ($p=0.11$). Larger tokenization change does *not* predict which examples flip.

6 Analysis and Discussion

Q1 — which method is most robust. RAG-augmented prompting. On the four rule-based, meaning-preserving families the RAG rows have the smallest Robustness Gaps (several slightly negative, i.e. the perturbation coincidentally helps) and the lowest Flip-Rates among the prompting methods. Retrieving semantically similar

Hebrew demonstrations appears to re-anchor the model on meaning rather than surface form, and this outweighs the retrieval drift introduced by perturbing the query.

The LoRA paradox. The Hebrew-native LoRA model behaves exactly as its tokenizer predicts on two families and exactly opposite on a third. It is *perfectly invariant* to spelling and niqqud (gap and flip both 0) because DictaBERT's WordPiece vocabulary and its niqqud handling absorb those edits before they reach the weights. But it has the **largest orthographic gap of any condition on typos (0.056) and by far the highest typo Flip-Rate (0.118)**. Final-form normalization (ג→י, מ→ם) produces character sequences the Hebrew WordPiece vocabulary never saw in training, so words split into unfamiliar sub-tokens the fine-tuned head cannot map. The naive hypothesis — “a Hebrew tokenizer makes the model robust everywhere” — is therefore false; the tokenizer helps precisely on the phenomena it was trained to normalize and hurts on the one it was not.

Q2 — Hebrew vs. English instructions. No consistent advantage. Comparing the paired HE/EN rows, Hebrew instructions help on some cells (RAG-HE has a lower prefix and typos gap than RAG-EN) and hurt on others (RAG-EN has higher clean F1). Claude's strong English instruction-tuning generalizes to Hebrew content well enough that the language of the *instruction* is a second-order effect relative to the *method* (retrieval vs. none).

Q3 — does tokenization explain fragility? No — and this is our most interesting negative result. Tokenizer fertility clearly *differs* by tokenizer (Table 3): the BPE tokenizer is catastrophically sensitive to niqqud while the Hebrew WordPiece tokenizer is immune. One might expect flips to track this. They do not (all $|r| < 0.05$, all $p > 0.1$). Two mechanisms reconcile this. First, the family with the largest fragmentation — niqqud under BPE — is not the family with the most flips; niqqud flips are among the *lowest*, because modern LLMs evidently treat vowel points as near-ignorable. Second, LoRA's fragility is concentrated on typos, where its own tokenization change is only modestly larger than average — so the failure is about *out-of-vocabulary sub-tokens*, not the raw count of tokens. Fragility is a property of *which* tokens appear, not *how many*.

Qualitatively, the flips are linguistically sensible failures rather than noise:

niqqud: Neutral → Positive — עונה על סקר על בירות וחושבת רק על בירה מכבי 7

typos: Neutral → Negative — היה יקר יותר וחיפשתי משהו עם פחות טבק

prefix: Positive → Neutral — יום טוב ו שבוע טוב

6.1 Limitations

- The spelling family is a high-precision curated rewrite, so its coverage is deliberately small (8.2%); its gap/flip are reliable but based on fewer examples.
- Niqqud insertion depends on the Dicta Nakdan service; when unavailable it degrades to zero coverage rather than corrupting the run.
- The colloquial family is LLM-generated and only sentiment-validated rewrites are kept; being a semantic edit it is reported separately from the four rule-based families.
- Claude's tokenizer is not public, so the tokenization-causality claim rests on Claude's measured input_tokens plus tiktoken as an open BPE reference. A future run with an open-tokenizer model (e.g. GPT-4o with o200k) would let the model's exact tokenizer drive the correlation.

7 Conclusion

Clean accuracy hides how differently Hebrew sentiment methods behave under meaning-preserving orthographic variation. Retrieval-augmented prompting is the most robust method overall; the Hebrew-native LoRA model is perfectly invariant to spelling and niqqud yet the most fragile of all to final-form typos, showing that a Hebrew tokenizer confers robustness only on the phenomena it was trained to normalize. Instruction language (Hebrew vs. English) is a minor factor next to method choice. Finally, although tokenizers differ enormously in how they fragment perturbed Hebrew, that fragmentation does not predict prediction flips: fragility is governed by *which* sub-tokens are produced, not how many. For Hebrew NLP practitioners the practical lesson is to evaluate on meaning-preserving variants, not just clean text, and to prefer methods (like RAG) that anchor on meaning over surface form.

AI-Tool Disclosure

Coding assistants were used for code scaffolding and prompt drafting. All experimental decisions, the evaluation design, and the analysis are our own, and every reported number is produced by code we can trace and re-run. Claude Haiku 4.5 is used as an evaluated system (the prompting methods) and to generate the sentiment-validated colloquial paraphrases.